SL Math School: Random Growth Models, etc.
Tutorial 7: Monotonicity and Stochastic Domination

1 Prologue

Fix two sets of boundary conditions: $(\mathbf{x}, \mathbf{y}, f, g)$ and $(\tilde{\mathbf{x}}, \tilde{\mathbf{y}}, \tilde{f}, \tilde{g})$, where $\mathbf{x}, \mathbf{y}, \tilde{\mathbf{x}}, \tilde{\mathbf{y}} \in \mathbb{R}^n_{\geq}$, and $f, g, \tilde{f}, \tilde{g} : I \to \mathbb{R}$, where $I \subset \mathbb{R}$ is a compact interval. Assume that

$$f(r) \geq g(r), \qquad \tilde{f}(r) \geq \tilde{g}(r)$$

for all $r \in I$. Consider two Dyson Brownian bridges, or melons,

$$B = (B_1, \ldots, B_n), \qquad \tilde{B} = (\tilde{B}_1, \ldots, \tilde{B}_n),$$

on the interval I with the corresponding boundary data, i.e.

$$B \sim \mathbb{P}_{\mathrm{NI}}^{I,\mathbf{x},\mathbf{y},f,g}, \qquad \tilde{B} \sim \mathbb{P}_{\mathrm{NI}}^{I,\tilde{\mathbf{x}},\tilde{\mathbf{y}},\tilde{f},\tilde{g}}.$$

Definition: We say that the boundary data $(\mathbf{x}, \mathbf{y}, f, g)$ is lower than the boundary data $(\tilde{\mathbf{x}}, \tilde{\mathbf{y}}, \tilde{f}, \tilde{g})$ if, for all i,

$$x_i \leq \tilde{x}_i, \qquad y_i \leq \tilde{y}_i,$$

and also

$$f(r) \leq \tilde{f}(r), \qquad g(r) \leq \tilde{g}(r)$$

for all $r \in I$. We denote this by

$$(\mathbf{x}, \mathbf{y}, f, g) \preceq (\tilde{\mathbf{x}}, \tilde{\mathbf{y}}, \tilde{f}, \tilde{g}).$$

Theorem: If $(\mathbf{x}, \mathbf{y}, f, g) \preceq (\tilde{\mathbf{x}}, \tilde{\mathbf{y}}, \tilde{f}, \tilde{g})$, then there exists a coupling of B and $\tilde{B}$ so that, for all $i = 1, \ldots, n$ and $r \in I$,

$$B_i(r) \leq \tilde{B}_i(r).$$

Note that in the above, the boundary data may be deterministic or random. In the random case, we should assume that the two sets of boundary data are coupled so that the ordering holds almost surely.

More precisely, there exists a single probability space $\mathbb{P}$ containing line ensembles

$$B \sim \mathbb{P}_{\mathrm{NI}}^{I,\mathbf{x},\mathbf{y},f,g}, \qquad \tilde{B} \sim \mathbb{P}_{\mathrm{NI}}^{I,\tilde{\mathbf{x}},\tilde{\mathbf{y}},\tilde{f},\tilde{g}},$$

so that, for all $i = 1, \ldots, n$ and $r \in I$,

$$B_i(r) \leq \tilde{B}_i(r).$$

The above statement is obvious in the Brownian bridge case, when $n = 1$, $f = \infty$, and $g = -\infty$, as one can simply let $\tilde{B}(r) = B(r) + L(r)$, where $L(r)$ is a non-negative affine linear function which corresponds to shifting the endpoints. For the general n case with arbitrary boundary data, one can apply a type of Markov chain or Metropolis algorithm for discrete random walks conditioned not to intersect, and then take a scaling limit.

2 Problems

2.1 Core Problems

Problem 1. Consider two sets of boundary data

$$(\mathbf{x}, \mathbf{y}, f, g) \preceq (\tilde{\mathbf{x}}, \tilde{\mathbf{y}}, \tilde{f}, \tilde{g}),$$

and two line ensembles

$$B = (B_1, \dots, B_n) \sim \mathbb{P}_{\mathrm{NI}}^{I,\mathbf{x},\mathbf{y},f,g}, \qquad \tilde{B} = (\tilde{B}_1, \dots, \tilde{B}_n) \sim \mathbb{P}_{\mathrm{NI}}^{I,\tilde{\mathbf{x}},\tilde{\mathbf{y}},\tilde{f},\tilde{g}}.$$

Show that, for all $i \in 1, \dots, n$, $t \in I$, and $a \in \mathbb{R}$,

$$\mathbb{P}\left(B_i(t) \geq a\right) \leq \mathbb{P}\left(\tilde{B}_i(t) \geq a\right).$$

Problem 2. Fix $T > 0$. For $n \in \mathbb{N}$, let W^n be an n-line Brownian melon on the interval $[-T, T]$, started and ended at $\mathbf{0} \in \mathbb{R}^n$. Show that there exists a coupling of W^n and W^{n-1} so that

$$W_1^n(x) \geq W_1^{n-1}(x) \qquad \forall x \in [-T, T].$$

That is, show that W^n and W^{n-1} can be defined on the same probability space so that the above condition holds.

Problem 3 (Concavity of Melon Shape)**.** Consider a Dyson n-Brownian bridge, or melon, $(W_1, \dots, W_n)$ on the interval $[-T, T]$ with fixed endpoints $W(-T), W(T) \in \mathbb{R}^n_>$.

Assume that W_1 is the top curve. Using the Brownian Gibbs property, show that the function

$$x \mapsto \mathbb{E}W_1(x)$$

is concave. That is, show that for all $s < t$ in $[-T, T]$ and all $\lambda \in (0, 1)$,

$$\mathbb{E}W_1\big(\lambda s + (1-\lambda)t\big) \geq \lambda \mathbb{E}W_1(s) + (1-\lambda)\mathbb{E}W_1(t).$$

2.2 Challenge Problem

Problem 4 (Increment bound for the Brownian melon)**.** Let $B = (B_1, \dots, B_n)$ be the n-curve Brownian melon on $[0, 1]$ with endpoints $B(0) = B(1) = \mathbf{0} \in \mathbb{R}^n$. Assume that the curves are ordered from top to bottom, so that B_1 is the top curve. Fix $i \in 1, \dots, n$ and $s \in [0, 1)$. Let $C = (C_1, \dots, C_{n-i+1})$ be an independent $(n-i+1)$-curve Dyson Brownian melon on the interval $[0, 1-s]$, and let $D = (D_1, \dots, D_i)$ be an independent i-curve Dyson Brownian melon on the interval $[0, 1]$, both with endpoints pinned at $\mathbf{0}$.

Prove that, for all $a > 0$ and $t \in (0, 1-s]$,

$$\begin{aligned}\mathbb{P}\left(B_i(s+t) - B_i(s) \geq a\right) &\leq \mathbb{P}\left(C_1(t) - \frac{D_i(s)t}{1-s} \geq a\right) \\ &\leq \mathbb{P}\left(C_1(t) \geq \frac{a}{2}\right) + \mathbb{P}\left(D_i(s) \leq -\frac{a(1-s)}{2t}\right).\end{aligned}$$

Hint: Condition on the ensemble at time s and apply the Brownian Gibbs property on $[s, 1]$. Use monotonicity to compare B_i from above by the top curve of a Brownian melon with $n-i+1$ curves, plus a linear function. Bound this endpoint correction using a comparison with the bottom curve of an i-curve melon.